Review Document

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1 Project Overview

1.1 Progress and Updates Summary

1.1.1 Overview

During the first phase of development, the Flask application and overall project structure were established. The application was configured with Flask routes and HTML templates, and the codebase was organized into functional blocks. Additional improvements included enhancing code organization through detailed comments and workflow documentation. The application was also transformed from a terminal-based program into a browser-based web application, providing a more interactive user experience.

By the end of Version 1, RecipeWise supports recipe searching, recipe details viewing, saving favorite recipes, and managing saved recipes through a browser-based interface.

1.1.2 Blocks

To keep the development process organized and easy to understand, RecipeWise was divided into functional blocks. Each block focuses on a specific feature of the application and builds upon functionality implemented in previous blocks.

- Block 1: Search Recipes
- Block 2: Recipe Details Page
- Block 3: Favorites Page
- Block 4: Persistent Storage with Supabase (Planned for Version 2)

The following sections summarize the progress, updates, and implementations completed within each block during Version 1 development.

Block #1: Search Recipes

Purpose:

Block #1 focused on creating the core recipe search functionality for RecipeWise. The objective was to connect the application to TheMealDB API and provide users with a browser-based interface for searching recipes.

Progress and Updates:

Development began by replacing the original terminal-based search experience with a Flask web interface. A search form was added to the Home Page (index.html) and connected to Flask

routes that process user searches. The application was integrated with TheMealDB API through the search_recipes() function, allowing recipe information to be retrieved dynamically.

Search results were enhanced by displaying recipe names, images, and country-of-origin information. Recipe names were also made clickable to support navigation to the Recipe Details page.

Outcome:

RecipeWise successfully allows users to search for recipes through a web interface and displays dynamic results retrieved from TheMealDB API.

Block #2: Recipe Details Page

Purpose:

Block #2 focused on enhancing the user experience by providing detailed information for individual recipes. Users can view ingredients, measurements, preparation instructions, and images for a selected recipe.

Progress and Updates:

A dedicated Recipe Details page (recipe_details.html) was created and connected to the search results through recipe-specific meal IDs. A second API request was implemented using TheMealDB lookup endpoint, enabling the retrieval of detailed recipe data for a selected recipe.

The Recipe Details page was expanded to display recipe names, images, ingredients, measurements, and preparation instructions. A Save Recipe button was added to support the Favorites functionality developed in Block #3. Additional improvements included formatting recipe instructions for improved readability and enhancing project documentation through comments and workflow descriptions.

Outcome:

Users can select a recipe from the search results and access a detailed view containing all the information required to prepare the recipe.

Block #3: Favorites Page

Purpose:

Block #3 focused on allowing users to save and manage recipes they wish to revisit later.

Progress and Updates:

A Favorites page (favorites.html) was created along with functionality for saving recipes from the Recipe Details page. Recipes are temporarily stored using an in-memory favorites list and

displayed with their associated information, including recipe name, image, and country of origin.

Additional functionality was implemented to improve usability, including clickable recipe names, duplicate-prevention logic, user feedback messages, and the ability to remove recipes from the favorites list.

Outcome:

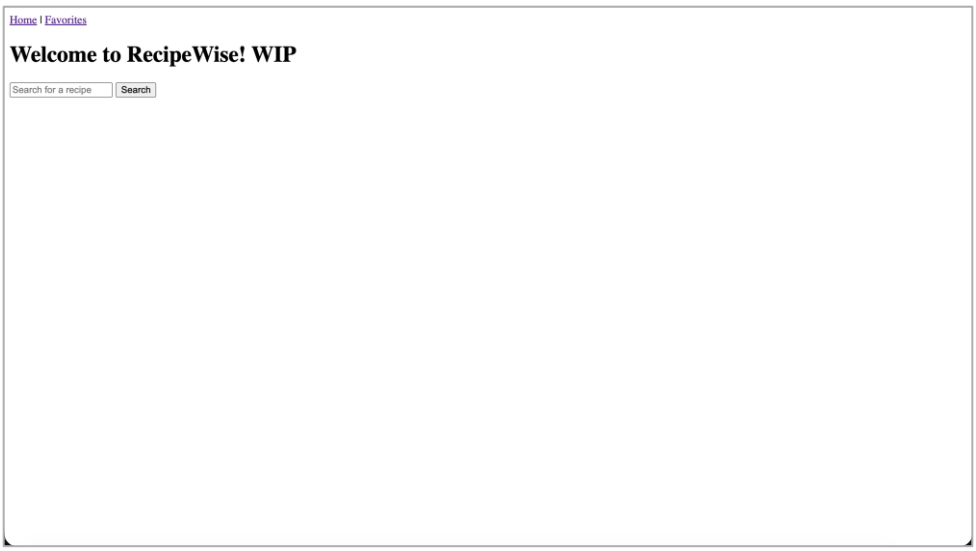
Users can build and manage a personalized collection of favorite recipes by saving, viewing, and removing recipes when needed.

1.2 Project Showcase

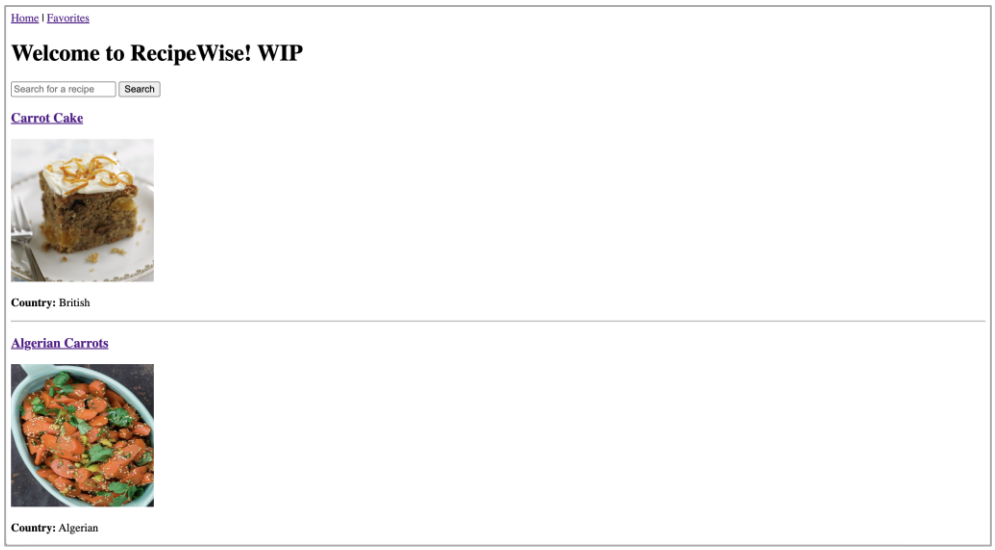
This section provides visual examples of the functionality implemented during Version 1 development.

Demonstration video (RecipeWise_V1 Summary.mov here) and screenshots bellow highlight the major features completed within Blocks 1, 2, and 3.

1.2.1 Block #1: Search Recipes



Home Page - Empty State



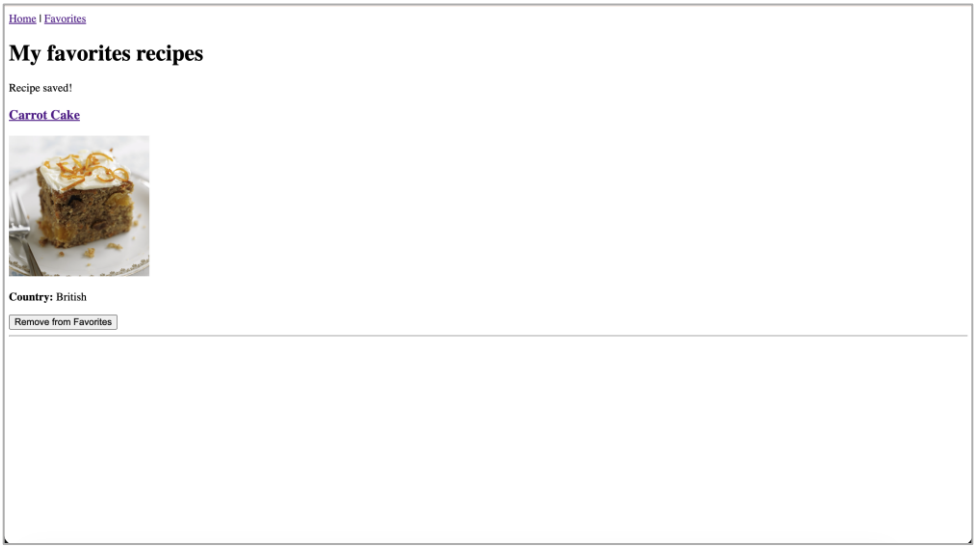
Home Page - Search Results List.

1.2.2 Block #2: Recipe Details Page

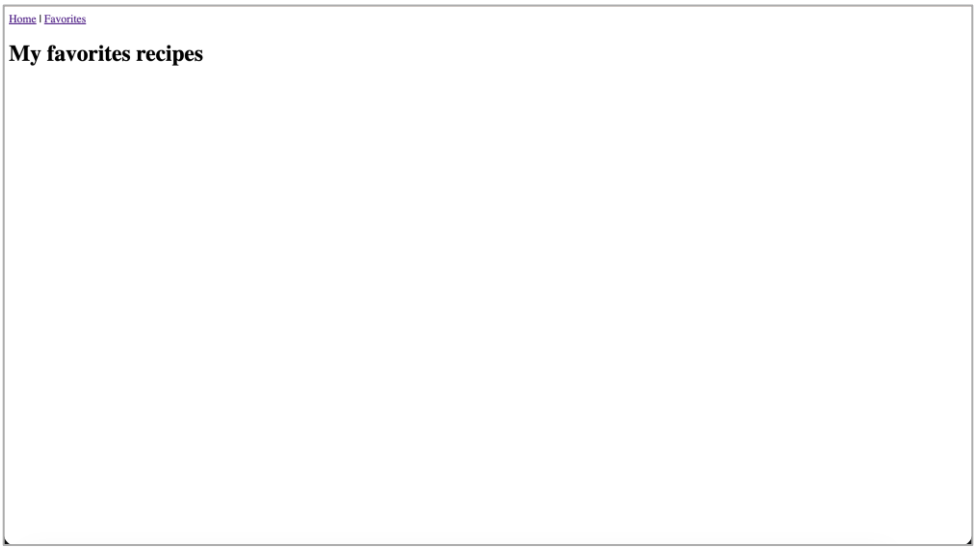


Recipe Detail Page with *<Add to Favorites>* button.

1.2.3 Block #3: Favorites Page



Favorite Page – Recipe Saved with *<Remove from Favorites>* button.



Favorites Page - Empty State

2 Milestones

2.1 Milestone 1: Implement Block #4

Replace the temporary in-memory favorites list with a Supabase database. Implement the ability to save, retrieve, and remove favorite recipes using persistent storage.

- Import and create Supabase project
- Connect Flask application to Supabase
- Save favorite recipes to the database
- Retrieve favorite recipes functionality
- Remove favorite functionality from the data base

2.2 Milestone 2: UI Redesign

Improve the visual appearance of RecipeWise by redesigning page layouts and improving readability. Actionable items:

- Improve spacing and alignment
- Improve page organization
- Make layouts more consistent across pages

2.3 Milestone 3: Visual Enhancements

Introduce visual components that make the application more engaging and professional. Actionable items:

- Add icons
- Create recipe cards
- Improve image presentation
- Add hover effects and visual feedback
- Improve button styling

2.4 Milestone 4: User Experience Improvements

Enhance usability by providing better feedback, error handling, and navigation. Actionable items:

- Empty favorites message
- Better success/error notifications
- Improved navigation between pages
- Better handling of invalid searches
- Improved accessibility

2.5 Milestone 5: Final Review and Polishing

Do a final clean up and make sure everything is working properly.

- Code cleanup
- Review comments and documentation
- Verify all workflows

- Final UI adjustments

2.6 Nice to have: Recipe Information Improvements

TheMealDB recipes do not follow a consistent instruction format. Some recipes use numbered steps, while others use paragraphs or different line-break structures. If time permits, future work may explore ways to standardize instruction formatting to improve readability across recipes.

3 Self-Reflection

Wins

Overall, I am very satisfied with the progress made during Version 1 of the project. During this phase, I successfully completed the three planned development blocks, which gives me confidence that the project is moving in the right direction. One of the most rewarding aspects of this version was seeing the application evolve from a simple terminal-based program into a browser-based web application with multiple pages and interactive features.

Dividing RecipeWise into functional blocks helped me focus on one feature at a time and made the project feel much less overwhelming. Instead of trying to build everything at once, I was able to approach development as a series of milestones. Looking back, this approach not only helped me stay organized but also allowed me to see steady progress and develop a better understanding of how the different parts of the project worked together.

Challenges

One of the biggest challenges I encountered was integrating Flask with TheMealDB API and understanding how information flows between Python and HTML templates. Moving from a terminal-based application to a web-based application required understanding how different components of the application communicate with one another. While these concepts initially felt challenging, once I understood the overall workflow and how the connection between the backend and frontend worked, I was able to apply the same approach across multiple pages, including the Home, Recipe Details, and Favorites pages.

Another challenge that remains unresolved involves the recipe instructions returned by the API. Because the recipes are created and formatted differently, the instruction fields are not standardized. Some recipes contain clearly separated steps, while others are stored as large blocks of text. Since the formatting originates from the API itself, creating a consistent presentation for all recipes is more complicated than I originally expected and may require additional work in the future.

Future

Looking ahead, I am most excited about the next phase of development, which will focus on improving the user interface and overall user experience. Interface design is one of the areas that interests me the most, so I am looking forward to learning more about creating a cleaner, more organized, and visually appealing application. I am especially excited to redesign the layout, improve readability, and introduce visual elements such as icons and recipe cards.

A piece that makes me nervous is integrating Supabase and implementing persistent storage. However, as it is the last block, I think I have a good margin of time to achieve it as the core functionality is already in place, which will allow me to focus on refining the application, improving the overall user experience, and adding long-term data storage. Overall, I feel confident that the remaining goals are achievable and that the next stage of development will be both challenging and rewarding.